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TREND-BASED MOCK PAPER

GATE Metallurgical Engineering

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Metallurgical Engineering
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

10% of 500:

(Numerical Answer Type — Enter the numerical value)

Section B: Metallurgical Engineering

Q2. [1-MCQ] [1 mark]

Rank of 3x3 zero matrix:

- (A) 0
 - (B) 1
 - (C) 2
 - (D) 3
-

Q3. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
 - (B) Continuous but not differentiable at 0
 - (C) Neither continuous nor differentiable
 - (D) Differentiable at 0 only
-

Q4. [1-MCQ] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q5. [1-MCQ] [1 mark]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q6. [1-NAT] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q7. [1-NAT] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q8. [1-MCQ] [1 mark]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q9. [1-MCQ] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q10. [1-MCQ] [1 mark]

$d/dx [\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q11. [1-MCQ] [1 mark]

$P(|Z| < 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q12. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q13. [1-MCQ] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q14. [1-MCQ] [1 mark]

If a 3x3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q15. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q16. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q17. [2-NAT] [2 marks]

Number of spanning trees in complete graph K_4 :

(Numerical Answer Type — Enter the numerical value)

Q18. [2-MSQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q19. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q20. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$

(D) $\sqrt{||u|| * ||v||}$

Q21. [2-NAT] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-MSQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q23. [2-NAT] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) $e^{-3/s}$

Q25. [2-NAT] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q26. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
- (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q27. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q28. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{(-\lambda)(1+z)}$

Q29. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q30. [2-MSQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q31. [2-NAT] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix}^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q32. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 \\ 0 \end{bmatrix}^T$
- (B) $\begin{bmatrix} 2 \\ 1 \end{bmatrix}^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q33. [1-MCQ] [1 mark]

BCC coordination number:

- (A) 4
- (B) 6
- (C) 8
- (D) 12

Q34. [1-NAT] [1 mark]

Phase diagram lever rule finds:

(Numerical Answer Type — Enter the numerical value)

Q35. [1-NAT] [1 mark]

BCC structure coordination number:

(Numerical Answer Type — Enter the numerical value)

Q36. [1-MCQ] [1 mark]

Eutectic reaction:

- (A) Liquid $\rightarrow$ α + β
- (B) α $\rightarrow$ β + γ
- (C) Liquid $\rightarrow$ α
- (D) α + β $\rightarrow$ γ

Q37. [2-MCQ] [2 marks]

Eutectic reaction:

- (A) Liquid $\rightarrow$ α + β

- (B) α & β + γ
- (C) Liquid & α
- (D) α + β & γ

Q38. [2-MCQ] [2 marks]

Phase diagram lever rule finds:

- (A) Temperature
- (B) Phase fraction
- (C) Composition
- (D) Density

Q39. [2-NAT] [2 marks]

Eutectic reaction:

(Numerical Answer Type — Enter the numerical value)

Q40. [2-NAT] [2 marks]

BCC structure coordination number:

(Numerical Answer Type — Enter the numerical value)

Q41. [2-MCQ] [2 marks]

Eutectic reaction:

- (A) Liquid & α + β
- (B) α & β + γ
- (C) Liquid & α
- (D) α + β & γ

Q42. [2-MCQ] [2 marks]

BCC structure coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q43. [2-MSQ] [2 marks]

BCC structure coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q44. [1-MCQ] [1 mark]

Steel hardening process:

- (A) Annealing
- (B) Quenching
- (C) Normalizing
- (D) Tempering

Q45. [1-MCQ] [1 mark]

Brass is:

- (A) Cu-Zn
- (B) Cu-Sn

- (C) Cu-Ni
- (D) Cu-Al

Q46. [1-MCQ] [1 mark]

FCC coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q47. [2-MCQ] [2 marks]

BCC coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q48. [2-MCQ] [2 marks]

Steel hardening process:

- (A) Annealing
- (B) Quenching
- (C) Normalizing
- (D) Tempering

Q49. [2-MCQ] [2 marks]

FCC coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q50. [2-MCQ] [2 marks]

FCC coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q51. [2-NAT] [2 marks]

Brass is:

(Numerical Answer Type — Enter the numerical value)

Q52. [2-NAT] [2 marks]

FCC coordination number:

(Numerical Answer Type — Enter the numerical value)

Q53. [1-NAT] [1 mark]

Reducing agent in blast furnace:

(Numerical Answer Type — Enter the numerical value)

Q54. [1-MCQ] [1 mark]

Blast furnace main product:

- (A) Steel
- (B) Pig iron
- (C) Copper
- (D) Aluminum

Q55. [1-MCQ] [1 mark]

Halls-Heroult process produces:

- (A) Iron
- (B) Aluminum
- (C) Copper
- (D) Zinc

Q56. [2-MSQ] [2 marks]

Halls-Heroult process produces:

- (A) Iron
- (B) Aluminum
- (C) Copper
- (D) Zinc

Q57. [2-MCQ] [2 marks]

Halls-Heroult process produces:

- (A) Iron
- (B) Aluminum
- (C) Copper
- (D) Zinc

Q58. [2-NAT] [2 marks]

Blast furnace main product:

(Numerical Answer Type — Enter the numerical value)

Q59. [2-MCQ] [2 marks]

Halls-Heroult process produces:

- (A) Iron
- (B) Aluminum
- (C) Copper
- (D) Zinc

Q60. [1-MCQ] [1 mark]

Edge dislocation moves by:

- (A) Climb
- (B) Glide
- (C) Cross slip
- (D) None

Q61. [1-MCQ] [1 mark]

Eutectic point has:

- (A) One phase
- (B) Two phases
- (C) Three phases simultaneously

(D) Four phases

Q62. [1-NAT] [1 mark]

Martensite is formed by:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-MCQ] [1 mark]

Cathodic reaction in rusting:

- (A) $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$
- (B) $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$
- (C) Both
- (D) None

Q64. [1-MCQ] [1 mark]

Shielded metal arc uses:

- (A) Gas shield
- (B) Flux coated electrode
- (C) Both
- (D) None

Q65. [1-MCQ] [1 mark]

Duralumin is Al +:

- (A) Cu
 - (B) Mg
 - (C) Si
 - (D) Mn
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1MCQ	A	1
3	1MCQ	C	1
4	1MCQ	B	1
5	1NAT	A	1
6	1MCQ	C	1
7	1NAT	B	1
8	1MCQ	B	1
9	1MCQ	B	1
10	1MCQ	A	1
11	1MCQ	B	1
12	1MCQ	B	1
13	1MCQ	B	1

14	1NAT	B	1
15	1NAT	B	1
16	1MCQ	B	1
17	1MCQ	A	1
18	1MCQ	A	1
19	1NAT	B	1
20	1NAT	A	1
21	1MCQ	A	1
22	1MCQ	B	1
23	1MCQ	A	1
24	1MCQ	A	1
25	1MCQ	C	1
26	1MCQ	A	1
27	1MCQ	A	1
28	1MCQ	C	1

29	1MCQ	A	1
30	1MCQ	C	1
31	2MCQ	C	2
32	2MCQ	B	2
33	2MCQ	B	2
34	2NAT	C	2
35	2MSQ	A	2
36	2MCQ	B	2
37	2MCQ	A	2
38	2NAT	A	2
39	2MSQ	C	2
40	2NAT	B	2
41	2MCQ	A	2
42	2NAT	A	2
43	2MCQ	A	2

44	2MCQ	A	2
45	2MCQ	A	2
46	2MCQ	B	2
47	2MCQ	A	2
48	2MCQ	B	2
49	2MCQ	B	2
50	2MSQ	A	2
51	2NAT	A	2
52	2NAT	B	2
53	2MSQ	B	2
54	2MCQ	C	2
55	2NAT	A	2
56	2MCQ	A	2
57	2MCQ	C	2
58	2MCQ	B	2

59	2MCQ	B	2
60	2MSQ	B	2
61	2NAT	B	2
62	2MCQ	A	2
63	2MCQ	B	2
64	2NAT	A	2
65	2NAT	C	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: easy

Q3.

Answer: C

Solution:

Correct Answer: C

Subject: Physical Metallurgy | Topic: Crystal Structure | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Mechanical Metallurgy | Topic: Dislocations | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Extractive Metallurgy | Topic: Iron Making | Difficulty: moderate

Q6.

Answer: C

Solution:

Correct Answer: C

Subject: Phase Transformations | Topic: Phase Diagrams | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Heat Treatment | Topic: Steel | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Corrosion | Topic: Electrochemical | Difficulty: easy

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Welding & Joining | Topic: Arc Welding | Difficulty: moderate

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Non-Ferrous Metals & Alloys | Topic: Aluminum | Difficulty: difficult

Q11.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q12.

Answer: B

Solution:

Blast furnace: iron ore + coke + limestone → pig iron (high carbon iron).

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate

Q13.

Answer: B

Solution:

Quenching: rapid cooling from austenitizing temp. Forms hard martensite.

Subject: Material Science | Topic: Material Science | Difficulty: difficult

Q14.

Answer: B

Solution:

Lever rule: fraction of phase = opposite arm / total tie-line length.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy

Q15.

Answer: B

Solution:

BCC: 8 nearest neighbors at cube corners.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: difficult

Q16.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q17.

Answer: A

Solution:

Brass: copper-zinc alloy. Bronze: copper-tin.

Subject: Material Science | Topic: Material Science | Difficulty: easy

Q18.

Answer: A

Solution:

Area = $\pi \cdot r^2$. Circumference = $2 \cdot \pi \cdot r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q19.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q20.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q21.

Answer: A

Solution:

$L\{t^2 f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q22.

Answer: B

Solution:

Hall-Heroult: electrolytic reduction of alumina (Al_2O_3) to aluminum.

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: easy

Q23.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q24.

Answer: A

Solution:

Eutectic: $L \rightarrow \alpha + \beta$ at eutectic temperature. One liquid to two solids.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: difficult

Q25.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q26.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q27.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4}$ approx 0.908.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q28.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q30.

Answer: C

Solution:

FCC: 12 nearest neighbors. Coordination number = 12.

Subject: Material Science | Topic: Material Science | Difficulty: moderate

Q31.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: B

Solution:

BCC: 8 nearest neighbors. Coordination number = 8.

Subject: Material Science | Topic: Material Science | Difficulty: easy

Q33.

Answer: B

Solution:

$$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q34.

Answer: C

Solution:

$$\text{Kirchhoff: for } K_n, \text{ spanning trees} = n^{n-2} = 4^2 = 16.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: A

Solution:

$$\text{Poisson: mean} = \lambda = \text{variance. Both} = 4.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: B

Solution:

$$f(0)=2, f(1)=0, f(-1)=4, f(2)=4. f(-2)=-8+6+2=0. \text{ Roots at } x=1 \text{ (double?) and } x=-2. f(x)=(x-1)^2(x+2). \text{ Two distinct real roots.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: A

Solution:

$$|\langle u, v \rangle| \leq \|u\| * \|v\|. \text{ Equality when } u \text{ and } v \text{ are linearly dependent.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q38.

Answer: A

Solution:

$$f(x)=x \text{ is odd on symmetric interval. Fourier series has only sine terms.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q39.

Answer: C

Solution:

$$y = A \cos(2x) + B \sin(2x). A=1, B=0. y(\pi/2) = \cos(\pi) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: B

Solution:

$$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: A

Solution:

$$L\{e^{-at}\} = 1/(s+a). \text{ Here } a=3: 1/(s+3).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: A

Solution:

$$f(i) = i^2 + 3i + 2 = -1 + 3i. \text{ Re}(f(i)) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q43.

Answer: A

Solution:

$$\text{Eigenvalues are } 3 \text{ and } 1. [1, 0]^T \text{ is eigenvector for eigenvalue } 1. A^{50} [1, 0]^T = [1, 0]^T.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q44.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

Eutectic: L $\rightarrow$ α + β at eutectic temperature. One liquid to two solids.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q48.

Answer: B

Solution:

Lever rule: fraction of phase = opposite arm / total tie-line length.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q49.

Answer: B

Solution:

Quenching: rapid cooling from austenitizing temp. Forms hard martensite.

Subject: Material Science | Topic: Material Science | Difficulty: moderate

Q50.

Answer: A

Solution:

$L\{f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q51.

Answer: A

Solution:

Eutectic: L $\rightarrow$ α + β at eutectic temperature. One liquid to two solids.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy

Q52.

Answer: B

Solution:

BCC: 8 nearest neighbors at cube corners.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy

Q53.

Answer: B

Solution:

Hall-Heroult: electrolytic reduction of alumina (Al_2O_3) to aluminum.

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate

Q54.

Answer: C

Solution:

FCC: 12 nearest neighbors. Coordination number = 12.

Subject: Material Science | Topic: Material Science | Difficulty: easy

Q55.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50} \cdot [1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: A

Solution:

Eutectic: L $\rightarrow$ alpha + beta at eutectic temperature. One liquid to two solids.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q57.

Answer: C

Solution:

FCC: 12 nearest neighbors. Coordination number = 12.

Subject: Material Science | Topic: Material Science | Difficulty: moderate

Q58.

Answer: B

Solution:

Hall-Heroult: electrolytic reduction of alumina (Al_2O_3) to aluminum.

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: easy

Q59.

Answer: B

Solution:

BCC: 8 nearest neighbors at cube corners.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy

Q60.

Answer: B

Solution:

BCC: 8 nearest neighbors at cube corners.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy

Q61.

Answer: B

Solution:

Blast furnace: iron ore + coke + limestone $\rightarrow$ pig iron (high carbon iron).

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: difficult

Q62.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50} \cdot [1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q63.

Answer: B

Solution:

Hall-Heroult: electrolytic reduction of alumina (Al_2O_3) to aluminum.

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: easy

Q64.

Answer: A

Solution:

Brass: copper-zinc alloy. Bronze: copper-tin.

Subject: Material Science | Topic: Material Science | Difficulty: moderate

Q65.

Answer: C

Solution:

FCC: 12 nearest neighbors. Coordination number = 12.

Subject: Material Science | Topic: Material Science | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 31 questions, 49 marks
Physical Metallurgy: 11 questions, 18 marks
Material Science: 9 questions, 15 marks
Extractive Metallurgy: 7 questions, 11 marks
General Aptitude: 1 questions, 1 marks
Mechanical Metallurgy: 1 questions, 1 marks
Phase Transformations: 1 questions, 1 marks
Heat Treatment: 1 questions, 1 marks
Corrosion: 1 questions, 1 marks
Welding & Joining: 1 questions, 1 marks
Non-Ferrous Metals & Alloys: 1 questions, 1 marks

Difficulty Distribution:

Easy: 20 | Moderate: 32 | Difficult: 13

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.